

AI/ML Capstone Project

D2 – Project Design Document (15% of final mark)

Due: Week 5 (softcopy on SLATE, hardcopy to your Professor in class)

Name 1: _____

Student Number 1: _____

Name 2: _____

Student Number 2: _____

Name 3: _____

Student Number 3: _____

Name 4: _____

Student Number 4: _____

Name 5: _____

Student Number 5: _____

Instructions:

- Please work on the following deliverable with your group.
- Prepare your deliverable report as discussed in class.
- Type your report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload your report in PDF format on SLATE. Team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stappled) of your deliverable to your Professor in class in **Week 5**.

Overall Objectives:

In this “**D2 – Project Design Document**”, the objective is to design a *realistic* and *achievable* project plan that defines technical architecture, milestones, and team deliverables using industry-standard practices. This document will serve as a foundational blueprint for all subsequent development phases, and it must be structured with clarity and professionalism.

Q1: Project Objectives

Create a **Work Breakdown Structure (WBS)** detailing technical tasks and deliverables. As part of your WBS, you will need to include at least **four Work Packages (WPs)**. These WPs should address the major objectives of the AI/ML system architecture and data flow you intend to work on in this capstone project. In the space below, provide an explicit heading for each WP followed by a brief description detailing what exactly you want to accomplish (in 4-6 sentences). [2 marks]

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Work Package 1: Multimodal Data Ingestion and Pre-processing

Our goal is to build an automated data pipeline that continuously collects diverse inputs for live traffic safety monitoring. We will write ingestion scripts to pull live text alerts directly from the Ontario 511 API and real-time weather. At the same time, the system will download live image feeds from Ministry of Transportation highway cameras alongside historical collision logs from the Toronto Police. To process the unstructured text, we will apply natural language processing techniques like tokenization to strip out the important words and isolate real road hazards. For the camera feeds, an automated OpenCV workflow will crop the frames and blur sensitive details like license plates or faces to maintain strict privacy standards.

Work Package 2: Predictive Machine Learning Model Development

This phase is focusing on building and training the intelligent core models that power our multi-layered risk analysis. We will design a Convolutional Neural Network using PyTorch to analyze highway camera frames and classify road conditions as dry, wet, snowy, or icy. For the incoming text data, a TF-IDF vectorizer will mathematically weight the severity of the language found in emergency alerts. We will then build a central Lasso-regularized Logistic Regression model to fuse these visual and text scores with live weather telemetry. Crucially, we will tune our training process to heavily maximize model recall, so the system prioritizes catching dangerous conditions over accuracy.

Work Package 3: Automate Backend Operations and Containerization

Goal here is to convert our machine learning scripts into a fully automated backend computing system. We will start by establishing a virtual environment with pinned dependencies, so the codebase runs identically on every team member's machine using GitHub to collaborate. Next, we will implement background automation routines that automatically fetch raw API data and run model predictions every few minutes. We will also serialize our trained neural network weights and configuration files so the backend can instantly load them during live cycles. Finally, the entire application runtime will be inside lightweight Docker containers to guarantee seamless deployment across different computer operating systems.

Work Package 4: Interactive Frontend Application UI

The final phase is to build an intuitive web application that turns complex backend predictions into clear safety insights for everyday drivers. We will begin using existing google map user interface to strategically arrange our warning cards, route suggestion, and map elements. Using the Streamlit framework, we will develop the live dashboard by binding our Python prediction endpoints directly to interactive front-facing sliders and buttons. The centerpiece of the interface will be a dynamic geographic map that automatically plots the user's intended driving path. This map module will actively update the color formatting of highway lines based on the specific hazard indices generated by our data fusion models.

Q2: Project Tasks

The focus here is on identifying the granularity of the WPs. For each WP, develop at least 3-4 tasks detailing the various subcomponents related to the AI/ML pipeline and applications. In the space below, provide a descriptive heading for each task. Then, for each labelled task, provide an explanation of what the task entails (in 1-2 sentences). Avoid being too redundant to what you indicated in Q1. [2 marks]

Work Package 1: Data Gathering & Preparation

- 1.1: Connecting to Live Data: We will write scripts to connect our system directly to the Ontario 511 and weather APIs, ensuring our data feed updates automatically.
- 1.2: Organizing Past Records: We will gather and clean the historical collision data from the Toronto Police, formatting it so our predictive models can easily learn from past trends.
- 1.3: Preparing Text: We will write text-processing rules to strip out filler words from emergency alerts, keeping only the critical terms that describe road hazards.
- 1.4: Blurring Privacy Info: We will set up an automated masking tool that blurs faces and license plates in the highway camera feeds to ensure strict public privacy.

Work Package 2: Building the AI Models

- 2.1: Training the Vision Model: We will train a neural network to look at highway photos and classify the physical road surface as dry, wet, snowy, or icy.
- 2.2: Scoring Alert Severity: We will build a text filter that reads the cleaned emergency alerts and assigns them a safety score based on the severity of the vocabulary.
- 2.3: Combining the Data: We will develop a fusion algorithm that mathematically blends the camera results, alert scores, and weather telemetry to calculate a final safety percentage.
- 2.4: Tuning for Safety: We will tune our model's settings to aggressively prioritize catching severe safety hazards, ensuring it warns drivers about danger rather than just optimizing for the shortest route.

Work Package 3: Setting Up the Backend

- 3.1: Standardizing Workspaces: We will create a shared software requirements list so every team member's computer runs the project exactly the same way without dependency crashes.
- 3.2: Setting Up Automation: We will write background scripts that allow the system to run on its own, automatically fetching new data and updating predictions every few minutes.
- 3.3: Saving Model Work: We will export our finished AI models into lightweight files so the backend can load them instantly during live operations.
- 3.4: Packing into Containers: We will package the entire application into containers, acting as portable software boxes that let the system run smoothly on any local machine.

Work Package 4: Designing the App Interface

- 4.1: Sketching the Layout: We will design the user interface around the Google Maps API, deciding exactly where the safety warnings and route cards will appear for the driver.
- 4.2: Building the Dash Dashboard: We will use the Dash library to build an interactive dashboard, including a custom slider that lets users evaluate safety scores across a specific range of coordinates.
- 4.3: Adding the Map Overlay: We will integrate a custom map layer that displays two to three route suggestions based on the user's origin, destination, and our live safety scores.

Q3: Task Owners

Assign roles and responsibilities for each team member. In other words, assign a “**task owner**” for each task within the WPs. This must be done *strategically* based on qualifications and interest of the team member being assigned a task. Common roles include, project manager, data engineer, AI/ML engineer, MLOps engineer, etc. Divide the workload equally among the team members so that it is fair, balanced, and realistic. In the space below, list the tasks from **Q2**, and indicate the name of the team member that will work on the specific task in parenthesis. Also, provide a specific description (in 1-2 sentences) why this specific team member is appropriate to work on the assigned task. [1 mark]

Q4: WBS Diagram

Using the descriptions in **Q1** (work packages), **Q2** (tasks), and **Q3** (task owner) organize your WBS to visually capture relevant information by way of a **tree diagram**. Your diagram should show the **hierarchical relationships** among all WPs, tasks, and it should include all relevant sub-branches. Also, make your diagram complete and coherent by enumerating the topics and by placing arrows to show all interconnected associations. Your diagram should visually display an overall summary for the workflow for your AI/ML project. You may use specialized software to make your diagram (e.g., Visio, draw.io, photoshop, etc.) [[1 mark](#)]

Q5: Weekly Milestones

Define **weekly milestones** and deliverables for project tracking. Your weekly milestones plan should cover Week 5 to Week 14 of the semester. Each week, you will need to complete a certain aspect of your project work (e.g., data gathering/cleaning, AI/ML modeling, model optimization, testing/prototyping, deployment/UI, tech. report writing, final presentation/demonstration). In the space below, enumerate the weeks from **Week 5 to Week 14**, and provide a **bulleted list** of the **tasks** and **sub-tasks** that you and your group aim to *realistically* accomplish by the end of each week. [2 marks]

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Q6: Gantt Chart

Using the results in **Q4** (WBS diagram) and **Q5** (weekly milestones), develop a detailed **Gantt chart**. The Gantt chart is a project management tool that visually illustrates a project schedule over time. Your Gantt chart should display a list of WPs/tasks vertically while being mapped to specific task owners. A timeline is shown horizontally using “horizontal bars” should be used to represent the needed duration in weeks. Make your Gantt chart as detailed as possible. Also, important AI/ML milestones should explicitly be indicated on your Gantt chart. An example of a Gantt chart that you can replicate is show in **Appendix A**. [\[2 marks\]](#)

Q7: Version Control

Establish **version control practices** and **repository management protocols** with **Git/GitHub**. This tool will allow you to track changes in source code and related files over the span of your AI/ML capstone project. This tool will also allow you to practically collaborate with team members without overwriting each other's work. In the space below, explain how you intend to use Git/GitHub for version control and for repository planning. Also describe in full detail the collaboration workflow and the strategy that you aim to apply. [1 mark]

Q8: Limitations, Constraints, and Risks

Identify **potential limitations, constraints, and risks** that you *may* face during this AI/ML capstone project. As you work on this question, please keep in mind that financial support from Sheridan to teams is not available, and that access to GPU-supported labs or clusters is not guaranteed. In the space below, develop a **risk mitigation table** that highlight the **top 5-7 risks** that you may encounter as you work on this project. Your table should also include a contingency plan for each risk. The contingency should propose a backup strategy to mitigate the potential risk, to avoid disruptions to the operations of your workflow, and to allow you to complete your AI/ML project. In the table below, provide a heading for each cell of the table that summarizes the risk and the related contingency. Then, provide a description for each cell of the table (in 2-3 sentences). [2 marks]

No.	Limitations, Constraints, Risks	Contingency Plan
#1		
#2		
#3		
#4		
#5		
#6		
#7		

Q9: Project Outcomes

By Week 14, all teams must prepare: (i) **D4**, a formal **technical report** detailing the comprehensive work on the AI/ML capstone project, (ii) **D5a, slide presentation** of the technical report, and (iii) **D5b, a working live demonstration of the AI/ML solution**.

Q9.1 – What do you anticipate the major outcomes of your capstone project will be?

Q9.2 – What type of AI/ML method will your solution consider (i.e., regression, classification, clustering, or possibly a combination of different methods)?

Q9.3 – What type of UI/system will you have to showcase a working demonstration of your AI/ML solution (e.g., MobileApp, cloud service, locally/publicly hosted URL, AI-powered physical system – IoT, drone, or robot, or possibly a combination of various UI/systems)?

In the space below, provide specific answers to these three questions to demonstrate that your anticipated capstone results are *thoroughly considered* and *strategically planned*. [2 marks]

References (Include 12-15 peer-reviewed sources to back your AI/ML capstone project ideas cited above with your answers. You may reuse references from “D1 – Project Proposal”).

[1] ?

[2] ?

[3] ?

[4] ?

[5] ?

[6] ?

[7] ?

[8] ?

[9] ?

[10] ?

[11] ?

[12] ?

Appendix A: Example of “Gantt Chart”

							Phase One												Phase Two								
WBS Number	Task Title	Task Owner	Start Date	Due Date	Duration	% of Task Complete	Week 1					Week 2				Week 3				Week 4					Week 5		
							M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	M
1	Project Conception & Initiation																										
1.1	Project Charter	Leon W	3/12/55	3/15/55	3	100%																					
1.1.1	Project Charter Revisions	Kylie R	3/15/55	3/16/55	1	100%																					
1.2	Research	Pete S	3/15/55	3/21/55	6	90%																					
1.3	Projections	Steve L	3/16/55	3/22/55	6	40%																					
1.4	Stakeholders	Allen W	3/17/55	3/22/55	5	70%																					
1.5	Guidelines	Malik M	3/18/55	3/22/55	4	40%																					
1.6	Project Initiation	Malik M	3/23/55	3/23/55	0	50%																					
2	Project Definition & Planning																										
2.1	Scope & Goal Setting	Steve L	3/24/55	3/28/55	4	5%																					
2.2	Budget	Allen W	3/29/55	4/2/55	4	30%																					

Good Luck!